

JYOTSHNA VANJARAPU

AI/ML ENGINEER • PYTHON DEVELOPER • DATA SCIENCE

Visakhapatnam, Andhra Pradesh, India | jyotshnavanjarapu@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI/ML-focused Python developer with a B.Tech in Computer Science and Engineering (AI/ML) and internship experience at HCLTech and Topnotch Software. Hands-on experience in Python backend development, REST APIs, PostgreSQL, machine learning, NLP, data preprocessing and model evaluation. Currently expanding into modern AI engineering with FastAPI, LLM applications, prompt engineering, embeddings, vector databases, RAG, LangChain, AI agents, LangGraph and MCP, while strengthening Data Science capabilities.

EXPERIENCE

Python Developer Intern | HCLTech Jul 2025 – Nov 2025

- Developed backend modules using Python and worked with structured data to support application data flow and backend operations.
- Worked with PostgreSQL for database schemas, data processing, query optimization and efficient backend data handling.
- Contributed to modular backend development with focus on maintainable code, data processing and performance optimization.

Python Developer Intern | Topnotch Software Private Limited Aug 2024 – Nov 2024

- Developed Python-based backend applications using Flask, implementing REST APIs and backend business logic.
- Applied machine learning and NLP techniques to a movie recommender system and Email/SMS spam classifier.
- Performed data preprocessing, model building and evaluation while debugging and improving application functionality.

SELECTED PROJECTS

Smart Attendance System | Python • OpenCV • Face Recognition • Flask • REST APIs

- Built a real-time face recognition and tracking system to automate attendance management.
- Developed a Flask backend for API requests and database interactions with an HTML/CSS interface.

Virtual Mouse Using Computer Vision | Python • OpenCV • Hand Tracking • Gesture Recognition

- Computer-vision project for touch-free computer interaction by translating tracked hand movements and gestures into mouse-control actions.
- Demonstrates real-time visual processing, hand tracking and event-driven interaction.

Movie Recommendation System & Email/SMS Spam Classifier | Python • ML • NLP • TF-IDF • Naive Bayes • Streamlit

- Implemented a content-based movie recommender using cosine similarity with an interactive Streamlit interface.
- Built an NLP-based spam classifier using Naive Bayes and TF-IDF with tokenization, stopword removal and stemming/lemmatization.

Upcoming Flagship: Agentic RAG & GenAI Platform | FastAPI • LLMs • Embeddings • Vector DB • RAG • LangChain • LangGraph • Agents • MCP

- Planned end-to-end GenAI application combining document ingestion, embeddings, vector retrieval, LLM reasoning and tool-using agentic workflows.
- Will demonstrate practical AI engineering across retrieval, context management, orchestration and tool execution.

TECHNICAL SKILLS

Programming: Python, SQL, C

Data Science & ML: Machine Learning, supervised learning, classification, recommendation systems, data preprocessing, feature engineering, model building, model evaluation

NLP: Natural Language Processing, tokenization, stopword removal, stemming, lemmatization, TF-IDF, Naive Bayes, text classification

Computer Vision: OpenCV, face recognition, image processing, real-time computer vision, hand tracking, gesture recognition

Backend & APIs: Flask, REST APIs, backend development, API integration, FastAPI — currently learning

Databases: PostgreSQL, MySQL, SQL, database schemas, query optimization

GenAI / LLM Engineering: LLMs, prompt engineering, embeddings, vector databases, RAG, LangChain, AI agents, LangGraph, MCP — currently learning

Tools: Git, GitHub, Jupyter Notebook, Streamlit, VS Code

EDUCATION

GMR Institute of Technology, Rajam 2021 – 2025

B.Tech, Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Sasi Educational Institute, Visakhapatnam 2019 – 2021

Higher Secondary Certificate | 95%

CERTIFICATIONS

Introduction to Databases — Coursera / Meta • Programming in Python — Coursera / Meta • Python for Data Science, AI & Development — Coursera / IBM